

Fig. 2: Semiconductor components distribution to end users

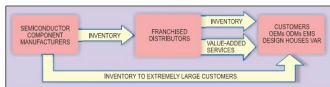


Fig. 3: Basic understanding of semiconductor distributors

width to augment the supplies when projects blossom.

**Labelling/packaging.** Under special agreements with customers, customised labelling and packaging can be offered which facilitates the manufacturing.

**Consolidated shipments and consolidated invoicing.** Normally, distributors avoid entertaining the cases involving components and products which they are not franchised for. As a special service, an agreement can be reached for clubbing the non-franchised items which can be either sourced by the distributor or arranged by the customer at distributor warehouse. A consolidated shipment is then made to the customer under a single invoice.

**Electronic data interchange (EDI).** Whilst distributors have systems to keep all their customers informed on the status of execution of their orders, they provide access to EDI facility to select customers. Customers can then track and monitor their order status in real time.

**E-commerce.** Many distributors have extended EDI for facilitating B2B and other e-commerce. In emergency, customers can go to a distributor's website on-line, purchase components using credit and debit cards besides electronic fund transfers.

**Quote cells.** Quote cells of distributors can be very helpful if involved right from the bidding stage. These quote cells are well equipped to even cross advise the parts when a non-franchised part is encountered in the parts list sent by the customer.

**Buffer stock management.** Despite the best management skills and

house rendering reduction in the cost of inventory carrying.

**Customer-managed inventory.** Using their strong warehouse management software and techniques, distributors can offer an upgraded service. Customers are able to pull the stocks allocated and locked specifically for them.

**Replenishment services.** As the name suggests, distributor refills a designated lot of inventory on getting a manual or automated notification. This is an implementation of Japan's famous Kanban concept of manufacturing in which production is done in the right size of batches.

**Lead time reduction services.** SCM team of customer works with the distributor to develop forecasting system. Using the forecast, distributor organises the availability of items on required dates of shipment. It is possible to attain shorter lead times after initial first shipment.

**Vendor-managed inventory (VMI).** Vendor-managed inventory is a continuous replenishment program in which the distributor manages and replenishes stocks at the customer store or warehouse level. This enables the distributor to project and anticipate the just enough amount of stock it needs to organise in order to meet the customer demand.

**Project implementation and safe passage.** In electronics field, product life span is short. New products and projects pop up very frequently. Every new project requires an implementation that is right for it. Distributors are apt at steering and navigating a safe and effective

implementation. Customers should call their intervention right in the beginning stage. More than OEMs, electronic manufacturing service (EMS) companies require this service.

**Expertise of servicing OEMs, ODMs and EMS.** Distributors do not restrict themselves to a very limited customer base. Their big and small customers fall into various categories, viz, OEMs, ODMs and EMS. They understand traits of each customer type in depth. Specialists for major accounts are appointed since the interaction style and service technique is diverse.

**Taping and reeling services.** Today's high-speed machines for electronics manufacturing require the feed lines to receive components in tape or reel form. Not always the components are available in desired packing. This means components need to be repacked in the required packing form.

**Programming and re-packing facilities.** Many components are one-time programmable (OTP) or multi-time programmable (MTF) and require to be loaded with software program before mounting on the boards. Distributors enter into agreements with their customers to do this activity at their end before shipping. They ensure that the devices are programmed properly, tested and checked before re-packing.

On customer's request, a copy of test report is also sent. Bar codes, labels, laser marking, holograms, quality and inspection certifications and stamps are also provided by distributors on the batches and consignment packs.

**Backward traceability.** Test data of all devices are kept by semiconductor manufacturers and distributors (for parts e-programmed by them) for a specified time for backward traceability, lest a quality issue arises. Also, many end customers have processes making it compulsory to comply. Distributors provide the customers test reports of components and products supplied by them on demand. Country of origin certificate (COC) is also supplied to customers on request.

**Rejection material acceptance (RMA).** Distributors and their princi-